

A Quieter Boat Propeller Could Help Whales Hear Each Other

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A Detroit company called Sharrow Marine is working with Ocean Alliance, a marine **conservation** organization, to test quieter boat **propellers** on research trips near whales. The tests are still at an early stage.

Noise pollution happens underwater as well as on busy streets. Large ships and other human-made machines add sound to the ocean. When the ocean is quiet, whales can hear each other's calls from hundreds of miles away. They use these calls to communicate and find food. Noise from shipping overlaps with the low sounds whales use, making their calls harder to hear.

Much of a boat's noise comes from its **propeller**, the spinning part that pushes it through water. A process called **cavitation** happens when tiny bubbles form behind a fast-spinning **propeller**. When the bubbles pop, they make hissing or crackling sounds.

Instead of blades with separate tips, Sharrow's design uses **looped** blades that twist like a ribbon. The company says this shape reduces **cavitation**. It claims independent testing found underwater noise reductions of up to seven **decibels** in the sound range whales use to communicate. **Decibels** are units used to measure sound levels. The company says results vary with operating conditions.

The idea began with a different problem. Greg Sharrow, the company's founder and CEO, wanted to film a violinist inside an orchestra using a camera **drone**. The **drone**'s buzzing would interfere with the music. He began thinking about a **propeller** without separate blade tips, then realized the idea could also work on boats. His early boat **propeller** tests took place in a fish tank.

Ocean Alliance CEO Iain Kerr supports solutions that benefit animals, the environment and boat users. The research trips will explore the quieter design near whales. Whether it meaningfully improves whales' ability to hear one another remains an open question.

Word Treasure

conservation -- Protecting nature and living things.

propeller -- A spinning part that pushes a boat through water.

cavitation -- The forming and popping of tiny bubbles near a propeller.

looped -- Shaped into a curve that joins back to itself.

decibels -- Units used to measure sound levels.

drone -- A flying machine with no pilot riding inside.

Background you need

Whales use calls to communicate and find food.

Sound levels can be measured in decibels.

The company says noise reductions vary with operating conditions.

Break it down

Introduces the research partnership and its early tests.

Explains why underwater noise matters to whales.

Describes cavitation and the looped propeller design.

Explains how a drone filming problem inspired the invention.

Ends with support for the idea and the unanswered question about whale communication.

Why it matters

Whales depend on hearing calls. Testing quieter boat parts could help researchers explore ways to reduce noise that interferes with those calls.

Test what you remember

1. What are Sharrow Marine and Ocean Alliance testing near whales?

- ☐ (A) Smaller underwater cameras
- ☐ (B) Quieter boat propellers
- ☐ (C) Lighter recording equipment
- ☐ (D) Faster research boats

2. Why can shipping noise cause problems for whales?

- ☐ (A) It makes the ocean water less salty.
- ☐ (B) It makes their calls harder to hear.
- ☐ (C) It makes their swimming harder to see.
- ☐ (D) It makes the ocean floor less rocky.

3. What makes hissing or crackling sounds during cavitation?

- ☐ (A) Tiny bubbles popping
- ☐ (B) Research boats turning
- ☐ (C) Looped blades bending
- ☐ (D) Camera drones landing

4. How is Sharrow's propeller shaped?

- ☐ (A) It has flat wheels.
- ☐ (B) It has square paddles.
- ☐ (C) It has straight tubes.
- ☐ (D) It has looped blades.

5. What first led Greg Sharrow to think about quieter propellers?

- ☐ (A) Measuring shipping noise near a port
- ☐ (B) Filming a violinist with a drone
- ☐ (C) Recording whale calls from a boat
- ☐ (D) Watching bubbles inside a fish tank

6. What does the article say about benefits for whale communication?

- ☐ (A) They end all shipping noise.
- ☐ (B) They replace the whales' calls.
- ☐ (C) They remain an open question.
- ☐ (D) They occur in every condition.

Answer key (try the quiz first!): 1: B 2: B 3: A 4: D 5: B 6: C

Questions to think about

1. **Greg Sharrow:** He says the propeller idea began with a noisy camera drone and could also be used on boats.
2. **Iain Kerr:** He supports solutions that benefit animals, the environment and boat users.

MY THOUGHTS (at least 20 words):